Mini-Project Milestone 1

Proposed Datasets for Pressure Prediction

DS675
Andres Arango Martinez
NJIT ID: 31748996 | UCID: aa3978

Area of interest and selection rationale. My research interests include pressure management in drinking-water distribution systems, particularly pressure behavior at remote critical nodes. I have previously worked on this problem using field measurements and model-based control methods, as described in a *published case study*. For this mini-project, I would like to study whether supervised machine learning can use current and historical SCADA measurements to forecast future pressure at a monitored node. The assignment asks for Kaggle datasets with usability above 9. BATADAL satisfies that criterion with a Kaggle usability of 10.0. BattLeDIM currently shows 3.53 on Kaggle, so it does not meet that specific threshold; I retain it as a candidate because its original source has substantial academic support, including a peer-reviewed ASCE benchmark paper, an official research archive, and EPANET network models.

Candidates at a glance

	BattLeDIM: L-Town	BATADAL: C-Town
Kaggle usability	3.53 (below the assignment threshold)	10.0 (meets the assignment threshold)
Kaggle link	Dataset page	Dataset page
Sampling	5 minutes	1 hour
Records	210,240 observations across 2018–2019	15,027 observations across three files
Main variables	33 pressures, 3 flows, 1 tank level, 82 AMR demands	12 pressures, 12 flows, 12 actuator states, 7 tank levels
Proposed task	Pressure 15 minutes ahead; supervised regression	Pressure 1 hour ahead; supervised regression

Candidate 1 BattLeDIM: L-Town

Dataset characteristics and academic provenance. BattLeDIM is a simulated SCADA benchmark for L-Town, a city-scale drinking-water network inspired by a real system in Cyprus. Its Kaggle copy has a usability rating of 3.53, which is below the assignment's requested value of greater than 9. I am including it transparently because the underlying dataset is not an informal Kaggle-only collection: it was developed for an international leakage-detection benchmark, documented in a peer-reviewed *Journal of Water Resources Planning and Management* paper, and released with an official Zenodo archive and EPANET network models. The benchmark paper describes 782 junctions and 905 pipe segments and divides the network into three hydraulic areas. The SCADA data contain 105,120 five-minute observations in each of 2018 and 2019, for 210,240 observations total. The main measurement table has 119 variables: 33 pressures, three flows, one tank level, and 82 automated meter reading (AMR) demand measurements. Leakage-event information is also available from the original benchmark for later diagnostic analysis.

Proposed predictive-learning task. I would use BattLeDIM for supervised regression: predict the recorded pressure head at one monitored node 15 minutes into the future. Predictors would be limited to information available at forecast time, such as current and lagged pressures, flows, tank level, AMR demands, and calendar features. I would select the target sensor after exploratory analysis of topology and pressure variability rather than labeling a node "critical" in advance. Because 119 candidate variables may be unnecessary, I would compare an all-network baseline with a topology-informed subset drawn from the same hydraulic area and its boundary connections. Leakage periods could later be used to evaluate whether forecast errors change under abnormal hydraulic conditions.

Sources: Kaggle | Official Zenodo archive | Vrachimis et al. (2022), ASCE

Mini-Project Milestone 1

Proposed Datasets for Pressure Prediction

DS675
Andres Arango Martinez
Candidate 2

Candidate 2 BATADAL: C-Town

Dataset characteristics and academic provenance. BATADAL is a simulated SCADA benchmark created for the Battle of the Attack Detection Algorithms on the C-Town water-distribution system. Its Kaggle copy has a usability rating of 10.0, satisfying the assignment requirement. Like BattLeDIM, it also has strong support beyond Kaggle: the original challenge provides official datasets, attack-event information, the C-Town EPANET model and control materials, and the benchmark was documented in a peer-reviewed ASCE paper. C-Town is a medium-sized network with 388 junctions, 429 pipes, seven tanks, 11 pumps, five valves, nine programmable logic controllers, and a SCADA system. The released data are sampled hourly and contain 43 process variables: seven tank levels, 12 pressures, 12 flows (11 pumps plus one actuated valve), and 12 binary actuator states. The three principal files contain 8,761 normal-operation records, 4,177 mixed development records, and 2,089 test records, for 15,027 observations total.

Proposed predictive-learning task. I would use BATADAL for supervised regression: predict the recorded pressure head at a selected monitored location one hour ahead. Candidate predictors would include current and lagged tank levels, flows, actuator states, pressures, and calendar/time features, but not attack labels or any future actuator information. The hourly sampling interval makes this a different forecasting problem from BattLeDIM and provides a useful comparison of how temporal resolution affects prediction. I would select the target node after reviewing network topology and pressure behavior and compare an all-network predictor set with a smaller hydraulically related subset, such as variables from the same district or from pumps, valves, tanks, and boundary flows that influence the target area. A later error analysis could separate normal and attack periods; because some BATADAL attacks deliberately falsify SCADA readings, that analysis would concern the predictability of the recorded signals rather than automatically treating them as true physical pressure.

Sources: Kaggle | Official BATADAL data | Taormina et al. (2018), ASCE

Planned evaluation in later stages. For either dataset, I would preserve chronological order when creating training, validation, and test periods, evaluate forecasts using regression metrics such as MAE and RMSE, and compare learned models against a simple persistence baseline (future pressure equals current pressure). This provides a clear supervised-learning framework while avoiding leakage of future information into model training.

Feature scope and dimensionality. The full measurement sets are useful starting points, but I would not assume that every network variable belongs in the final model. After choosing the target sensor, the EPANET topology can define a hydraulically related subset—for example nearby pressures, inlet or boundary flows, connected tank levels, relevant pump or valve states, and local demand information where available. Comparing this physically motivated subset with an all-network baseline makes feature reduction a testable modeling decision rather than an arbitrary preprocessing step.

Why I am considering these two datasets

Both benchmarks are directly related to pressure behavior in water-distribution systems, contain substantially more observations than candidate predictors, and include engineering context that can help interpret model inputs and errors. BATADAL fully meets the stated Kaggle-usability requirement, while BattLeDIM does not; BattLeDIM remains scientifically attractive because of the quality and reproducibility of its original academic benchmark. Together they frame a clear predictive question for the later stages: how accurately can future pressure be forecast from information that would actually be available at prediction time?